

Notes: Strömberg (QJE, 2004)

Radio's Impact on Public Spending

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1 Big picture

- **Question.** Does access to mass media change government policy by changing which voters are informed and electorally valuable?
- **Historical setting.** The paper studies U.S. counties during the early 1930s, when radio ownership was expanding rapidly and unevenly across space, and when the Federal Emergency Relief Administration (FERA) was distributing large New Deal relief funds.
- **Core mechanism.** Politicians target transfers to groups that can reward or punish them. Voters with radios are more likely to know who is responsible for relief spending and are more likely to vote, so governors have stronger incentives to direct funds toward counties with more radios.
- **Main empirical result.** Counties with higher radio penetration received more FERA relief spending per capita, even after controlling for unemployment, income proxies, demographics, voter turnout, state fixed effects, and other political variables.
- **Magnitude.** In the baseline OLS estimates, the elasticity of FERA spending with respect to the share of households with radios is about 0.14. At the mean radio share, a one percentage point increase in radio penetration implies roughly 0.54 percent higher relief spending.
- **Turnout channel.** Radio also increased voter turnout. The paper decomposes the total effect of radio on spending into a large direct information/accountability component and a smaller indirect turnout component.
- **Identification strategy.** The main cross-county estimates are supported by instruments based on AM radio propagation: ground conductivity and woodland share. These affect radio reception quality but are plausibly unrelated to the political demand for relief, conditional on controls.
- **Political economy takeaway.** The expansion of the franchise is not enough to equalize political influence. Voters also need information about political responsibility. The paper therefore argues for an “informed franchise” view of redistribution and government growth.

2 Incremental contribution of the paper

- **Brings mass media into distributive politics.** The paper moves the political-economy literature beyond voter income, ideology, and turnout by showing that media access itself changes the distribution of public spending.
- **Connects a formal model to a historical natural experiment.** The model predicts that spending rises with media access and turnout. The empirical setting - radio diffusion during the FERA program - provides unusually large cross-county variation to test those predictions.
- **Quantifies the policy impact of information technology.** Rather than only showing that media changes voting or beliefs, the paper estimates an effect on real fiscal allocations: New Deal relief dollars.
- **Separates direct information effects from turnout effects.** Radio matters partly because it increases turnout, but the larger channel is that voters with radios better know who is responsible for relief. This distinction is central to the paper’s contribution.
- **Introduces technology/geography instruments for media access.** Ground conductivity and woodland share generate variation in AM radio reception quality. This strategy anticipates later empirical work using geography or technology constraints to identify media effects.
- **Provides a historical template for studying new media.** The paper uses radio to study how communication technologies reallocate political power. The same logic can be applied to television, the internet, and social media.

3 Historical background and institutional setting

3.1 Radio in the early 1930s

Radio became an important political medium before and during the New Deal. By the early 1930s, political campaigns and news broadcasters used radio heavily, and radio was widely perceived as credible. But access was highly uneven: in the paper’s county data, the share of households with radios ranges from about 1 percent to 90 percent, with a mean around 26 percent.

Detailed interpretation: Radio is attractive for identification because it was politically important but not uniformly available. If informed voters get more favorable policy, the uneven geographic spread of radio should show up in the allocation of public funds.

3.2 The FERA program

The Federal Emergency Relief Administration operated from 1933 to 1935. It was a large relief program, distributing billions of dollars at a time when total government spending was much smaller than today. Relief was administered through a federal-state-county structure, with governors and state administrations playing a central role in allocating funds across counties.

Detailed interpretation: FERA is useful empirically because it generated large county-level transfers and because governors had enough discretion that political incentives could plausibly affect allocations.

3.3 Why radio could matter for FERA allocations

Radio could affect FERA spending in two ways:

- **Information/accountability:** voters with radios are more likely to know that the governor is responsible for relief allocations.
- **Turnout:** voters with radios are better informed and may be more likely to vote.

Detailed interpretation: Both channels raise the electoral value of relief spending in counties with high radio penetration. The paper’s empirical design estimates both the direct effect on spending and the indirect turnout channel.

4 Data and measurement

4.1 Unit of observation and main sample

The main empirical analysis is a cross section of U.S. counties. The dependent variable is cumulative FERA spending per capita from April 1933 to December 1935. The baseline OLS table uses 2,492 county observations after excluding observations with missing variables and including state fixed effects.

Detailed interpretation: The county is the right unit because FERA spending and radio ownership are both observed at the county level, and counties are the relevant local units receiving relief.

4.2 Main outcome: FERA spending per capita

The main spending variable is:

$$z_c = \frac{\text{cumulative FERA disbursement to county } c}{\text{county population}}.$$

Table I reports a mean of about \$20 per capita, with a standard deviation of about \$15 and a maximum above \$225.

Detailed interpretation: The large dispersion in FERA spending makes it possible to ask whether information and electoral variables help explain more than need-based variables alone.

4.3 Media access and political information

The key media variable is:

$$r_c = \text{share of households in county } c \text{ with radios in 1930.}$$

The paper also uses illiteracy and school enrollment as additional proxies for political information and ability to process political news. Radio is predicted to increase the probability that voters know who is responsible for relief spending.

Detailed interpretation: Radio is not just a consumer durable in the theory. It is a device that changes the political information set of voters.

4.4 Political variables

The main political variables are:

- **voter turnout:** votes per capita in the relevant gubernatorial election;
- **marginal voter density:** the historical volatility of Democratic vote shares in gubernatorial elections, used as a proxy for the density of swing voters.

Detailed interpretation: The model predicts that governors should spend more where more voters turn out and where there are more voters on the margin of switching support.

4.5 Need and socioeconomic controls

The regressions include controls for the need for relief and for economic conditions:

- unemployment in 1930 and 1937;
- bank deposits per capita and changes in bank deposits;
- median dwelling values, farm values, retail wages, crop values, and rents;
- age, gender, race, immigrant share, partisanship, urban/rural status, population density, and population.

Detailed interpretation: These controls are crucial because radio ownership is positively correlated with economic development. Without them, one might confuse media effects with differences in poverty, wealth, or political influence.

4.6 Instruments for radio ownership

The IV strategy uses geological and geographic determinants of AM radio reception:

- **ground conductivity** in the county seat;
- **share of county land area that is woodland**.

AM signals travel through both the ground and air. Higher ground conductivity improves AM signal propagation, while woodland can obstruct or alter reception quality.

Detailed interpretation: These instruments aim to isolate variation in radio penetration that comes from reception technology rather than from county demand for radios or county political conditions.

5 Model and empirical specifications

5.1 Governor's allocation problem

An incumbent governor allocates the state relief budget across counties:

$$\sum_c n_c z_c = I_s,$$

where n_c is county population, z_c is per capita relief spending, and I_s is the state budget.

Voters in county c derive utility from relief:

$$u_c(z_c) = k + \frac{a_c}{1 - 1/\alpha} z_c^{1-1/\alpha}, \quad a_c > 0, \quad 0 < \alpha < 1.$$

Detailed interpretation: The parameter a_c captures need: relief is more valuable in counties where economic hardship is more severe.

5.2 Political information and voting

A voter supports the incumbent only if the voter knows the incumbent is responsible for relief and if the resulting utility is high enough. Let $\xi_i = 1$ if voter i knows the governor is responsible for the relief allocation. The voting rule is:

$$\xi_i u_c(z_c) - \beta_i - \eta \geq \bar{u}_i.$$

The governor believes the probability of a voter being informed is:

$$\sigma_c(r_c) = \bar{\sigma}_c r_c^{c\sigma_r},$$

where r_c is radio penetration.

Detailed interpretation: Radio affects policy because it raises σ_c , the probability that voters connect relief spending to the responsible incumbent.

5.3 Optimal allocation

The governor maximizes reelection probability. The first-order condition for an interior allocation is:

$$t_c f_c \sigma_c u'_c(z_c) = \lambda_s,$$

where t_c is turnout, f_c is marginal voter density, and λ_s is a state-level shadow value of funds.

Taking logs yields:

$$\ln z_c = \alpha c_{\sigma r} \ln r_c + \alpha \ln \bar{\sigma}_c + \alpha \ln t_c + \alpha \ln f_c + \alpha \ln a_c - \alpha \ln \lambda_s.$$

Detailed interpretation: Counties receive more relief when voters are more informed, more likely to vote, more electorally marginal, and more in need of relief.

5.4 Baseline empirical equation

The estimable allocation equation is:

$$\ln z_c = c_1 \ln r_c + c_2 \ln t_c + \beta_1' x_{c1} + \mu_s + \varepsilon_{c1}.$$

Here:

- z_c is FERA spending per capita;
- r_c is radio penetration;
- t_c is voter turnout;
- x_{c1} contains information, need, political, and demographic controls;
- μ_s are state fixed effects.

The main predictions are:

$$c_1 > 0, \quad 0 < c_2 < 1.$$

Detailed interpretation: A positive c_1 means that media access changes the spatial allocation of public funds. A positive c_2 means that politicians favor counties where more people vote.

5.5 Turnout equation

The turnout specification uses a county panel from 1920 to 1930:

$$t_{ct} = b_1 r_{ct} + X_{ct}' \beta_2 + \mu_c + \mu_t + \varepsilon_{ct2}.$$

Detailed interpretation: This specification tests the indirect channel: radio may increase public spending partly by increasing voter turnout.

6 Identification and empirical design

6.1 Baseline cross-county comparison

The baseline design compares counties within the same state. State fixed effects absorb state budgets, state institutions, governors, and statewide relief-policy differences. The identifying variation is cross-county variation in radio penetration, voter turnout, and controls within states.

Detailed interpretation: This design is stronger than a national cross-sectional comparison because it holds constant state-level political institutions and the total state relief budget.

6.2 Main concern: omitted variables

Radio ownership is not randomly assigned. Counties with more radios may also be richer, more literate, more urban, more politically organized, or less in need of relief. The bias could go either direction:

- if richer counties have more radios but need less relief, OLS may be biased downward;
- if politically influential counties have more radios and more relief, OLS may be biased upward.

Detailed interpretation: The paper addresses this with a broad control set, state fixed effects, subsample tests, robustness checks, and IV estimates.

6.3 Instrumental variables design

Radio penetration is instrumented by ground conductivity and woodland share. The first stage is:

$$\ln r_c = \pi_1 \text{GroundConductivity}_c + \pi_2 \text{WoodlandShare}_c + X'_c \gamma + \mu_s + u_c.$$

The second stage replaces $\ln r_c$ with predicted radio penetration:

$$\ln z_c = c_1 \widehat{\ln r_c} + c_2 \ln t_c + X'_c \beta + \mu_s + \varepsilon_c.$$

Detailed interpretation: The exclusion restriction is that ground conductivity and woodland affect FERA spending only through radio penetration, not through unobserved need for relief or political influence.

6.4 Testing the turnout channel

The paper estimates the effect of radio on turnout using county-level panel variation between 1920 and 1930. County fixed effects absorb permanent county political culture, while election-year effects absorb common shocks to turnout.

Detailed interpretation: This design asks whether counties that gained more radio access experienced larger increases in votes per capita.

6.5 Subsample logic: competitive states and rural counties

The model predicts larger effects in places where electoral incentives or media information should matter more:

- **competitive states:** governors have stronger incentives to target marginal voters;
- **rural counties:** radio provided a bigger information improvement because alternative news sources, especially newspapers, were less available.

Detailed interpretation: The stronger estimates in competitive states and rural counties support the mechanism rather than a generic correlation between radios and county prosperity.

6.6 Robustness and placebo logic

The paper checks whether radio is merely proxying for consumer modernity or wealth by including gasoline sales per capita, which should be correlated with car ownership and consumer durables. Gasoline sales does not explain FERA spending in the same way. Results are also robust to restricting the sample by population size, weighting by population, and using robust or LAD-style estimators.

Detailed interpretation: These checks reduce concern that radio is simply standing in for rich or modern counties.

6.7 Bottom line on identification

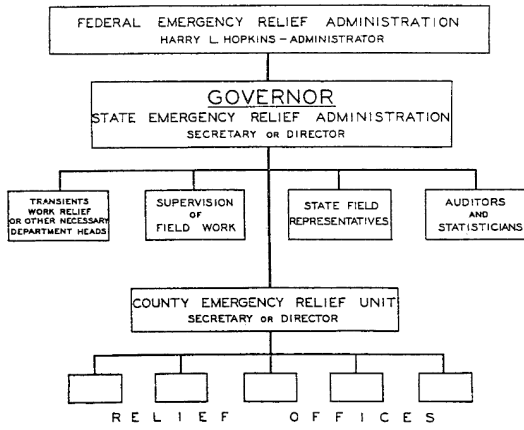
The paper is not a randomized experiment. The strongest evidence comes from the alignment of several pieces:

1. radio penetration predicts more FERA spending conditional on many controls;
2. voter turnout also predicts spending;
3. radio predicts turnout in panel data;
4. effects are stronger where the theory predicts, especially rural counties and competitive states;
5. IV estimates using reception-quality instruments are generally larger than OLS, especially in rural counties.

Detailed interpretation: The empirical case rests on mechanism consistency. The same informational theory predicts spending, turnout, heterogeneity, and the decomposition in Figure II.

7 Tables and figures

7.1 Figure I: FERA Organization Chart and Table I: Summary Statistics



(a) **Figure I: FERA Organization Chart.** The diagram shows the administrative hierarchy from the federal FERA office through governors and state emergency relief administrations to county relief units. It matters because the paper treats governors as central political actors in allocating relief across counties.

TABLE I
SUMMARY STATISTICS

Theoretical variables	Empirical variables	Mean	St. dev.	Min.	Max.
$z_c =$	FERA spending/capita	19.98	15.32	0.04	225.67
$\sigma_c = +$	share hhlds with radios	0.26	0.18	0.01	0.90
$-$	share illiterate	0.04	0.04	0.00	0.44
$+$	school enrollment	0.74	0.06	0.05	1.00
$t_c = +$	voter turnout	0.30	0.17	0.00	0.82
$f_c = +$	marginal voter density	0.09	0.06	0.00	0.31
$a_c = +$	unempl. 1930	0.01	0.01	0.00	0.09
$+$	unempl. 1937	0.04	0.02	0.00	0.14
$-$	bank deposits/capita	115	169	0.45	5345
$-$	%Δbank deposits/capita	-0.25	0.52	-1.00	11.61
$-$	median dwell. value	2582	1357	536	20000
$-$	farm value/capita	189	148	0.03	849
$-$	retail wage	1130	205	333	2800
$-$	crop value/capita	137	117	0.01	1272
$+$	median rent	1444	683	429	5204
$-$	share 21+	0.55	0.06	0.36	0.83
$+$	share 65+	0.04	0.02	0.00	0.09
$+$	share female	0.48	0.02	0.23	0.54
$+$	share black	0.11	0.18	0.00	0.86
$+$	share immigrants	0.05	0.06	0.00	0.50
controls	share partisans	0.64	0.23	0.00	1.00
	share urban	0.21	0.26	0.00	1.00
	rural dummy	0.44	0.50	0	1
	gas sales/capita	10	8	0	122
	population	40609	138268	48	4014611
	pop. density	1846	19341	1	848778
instruments	ground conductivity	9.27	8.13	0.5	30
	share woodland	0.06	0.08	0.00	0.42

(b) **Table I: Summary Statistics.** The table maps theoretical variables to empirical measures and reports dispersion across counties. It shows large variation in FERA spending, radio penetration, voter turnout, and economic need, which is what makes the empirical design informative.

Read:

- Figure I clarifies why the paper focuses on governors: relief funds passed through state administrations before reaching counties.
- Table I shows mean FERA spending of about \$20 per capita and mean radio penetration of 26 percent, with a wide range from 1 percent to 90 percent.
- The same table shows that turnout, illiteracy, school enrollment, and unemployment vary substantially across counties.

Detailed interpretation: Figure I motivates the political agency channel. Table I shows that the treatment variable is not marginal: radio access varied enormously in the early 1930s. This creates enough cross-sectional variation for radio to have detectable political consequences.

How it fits the paper: The visual pair links institutions to measurement. The organization chart tells us who had discretion; the summary table tells us which county characteristics explain how discretion was used.

7.2 Table II: OLS Estimates, Dependent Variable: Log FERA Spending/Capita

Read:

- The baseline coefficient on log radio penetration is 0.138 and significant at the 1 percent level.
- The baseline coefficient on log voter turnout is 0.165 and also significant at the 1 percent level.
- The radio coefficient rises to 0.264 in competitive states and to 0.201 in rural counties.
- Illiteracy is negative and school enrollment is positive, consistent with the broader information mechanism.
- Economic need variables also matter: unemployment in 1937 is strongly positively related to FERA spending.

Detailed interpretation: The table supports the paper’s two central empirical predictions: politicians allocate more relief to counties where more voters have access to political information and where more people vote. The bigger radio coefficient in rural counties is especially important because radio likely provided a larger information shock in areas with weaker newspaper access.

How it fits the paper: Table II is the paper’s core reduced-form evidence. It shows that information variables explain relief allocations even after conditioning on need, income proxies, demographics, and state fixed effects.

7.3 Table III: IV Estimates, Dependent Variable: Log FERA Spending/Capita

Read:

- The IV radio coefficient is 0.238 in the full sample, 0.617 in competitive states, and 0.717 in rural counties.
- The rural IV coefficient is statistically significant and much larger than the OLS estimate.
- First-stage F-statistics are 36, 22, and 22 for the radio specifications, so the instruments are not weak by usual standards.
- Overidentification tests do not reject instrument validity in the reported specifications.
- Instrumenting turnout produces a similar spending equation, suggesting that reverse causality from FERA spending to turnout is not driving the main spending result.

Detailed interpretation: The IV estimates suggest that OLS may understate the effect of radio, especially in rural counties. This is consistent with a downward omitted-variable bias: counties with more radios were often richer and therefore may have had less need for relief. The IV does not prove the exclusion restriction, but it strengthens the argument that radio access itself mattered.

How it fits the paper: Table III is the main causal-design table. It is the paper’s attempt to isolate variation in radio driven by AM reception technology rather than by socioeconomic or political county traits.

TABLE II
OLS ESTIMATES, DEPENDENT VARIABLE: LOG FERA SPENDING/CAPITA

A	B	I	II	III	IV
σ_c	+ c_1 : log share hhlds with radios	0.138** (2.6) (-1.11)*	0.145** (3.0) (-1.216)*	0.264** (3.8) (-2.133)	0.201** (3.6) (-1.577)**
	- share illiterate				
	+ school enrollment	0.856* (2.3)	0.789* (2.3)	.877 (1.6)	0.847 (1.9)
t_c	+ c_2 : log voter turnout	0.165** (2.9)	0.189** (3.3)	0.389** (2.9)	0.120* (2.4)
f_c	+ marginal voter density	0.034 (0.1)	0.137 (0.4)	0.077 (0.1)	-0.185 (-0.4)
a_c	+ unempl. 1930	7.837** (3.9)	6.493** (3.2)	8.449** (3.7)	0.088 (0.03)
	+ unempl. 1937	9.750** (10.6)	9.153** (10.6)	10.165** (6.5)	8.248** (5.6)
	- log bank deposits/capita	-0.093** (-3.6)	-0.088** (-3.2)	-0.125** (-3.0)	-0.122** (-2.9)
	- %bank deposits/capita	-0.013 (-0.9)		-0.116 (-1.8)	-0.009 (-0.9)
	- log median dwell. value	-0.0004 (-0.01)		-0.043 (-0.6)	-0.002 (-0.03)
	- log farm value/capita	-0.144* (-2.5)	-0.151** (-2.6)	-0.261** (-3.4)	-0.128 (-1.8)
	- log retail wage	0.016 (0.2)	-0.092 (-1.1)	-0.157 (-1.2)	-0.014 (-0.1)
	- log crop value/capita	0.017 (0.5)	.008 (0.2)	.093 (1.8)	.033 (0.9)
	+ log median rent	-0.063 (-0.9)		-0.029 (-0.2)	-0.034 (-0.3)
	- share 21+	-1.908** (-2.7)	-2.053** (-3.3)	-2.501* (-2.6)	-3.189** (-3.3)
	+ share 65+	-2.181 (-1.0)	-2.525 (-1.2)	-1.075 (-0.4)	1.936 (0.6)
	+ share female	1.923* (2.2)	1.760* (2.0)	1.016 (0.8)	0.988 (0.4)
	+ share black	0.105 (0.9)	0.156 (1.4)	-0.317 (-0.9)	.061 (0.4)
	+ share immigrants	0.319 (0.5)	0.207 (0.4)	-0.035 (-0.1)	0.066 (0.1)
control	+ share partisans	0.052 (0.2)	0.085 (0.5)	0.194 (0.8)	-0.011 (-0.04)
	share urban	0.994** (5.8)	1.002** (5.9)	1.008** (4.3)	
	rural dummy	0.253** (5.8)	0.257** (5.9)	.263** (4.3)	
	log gas sales/capita	0.015 (0.6)		-0.009 (-0.3)	0.021 (0.8)
	log pop. density	-0.064* (-2.4)	-0.069* (-2.3)	-0.087* (-2.4)	-0.042 (-0.7)
	log population	-0.092* (-2.4)	-0.106** (-2.7)	-0.020 (-0.5)	-0.302 (-5.4)
	state effects	yes	yes	yes	yes
	R^2	0.63	0.63	0.59	0.69
	Number of observations	2492	2679	1749	984
	p -value, H_0 : c_1 subsample =				
	c_1 full sample - subsample			0.00	0.08
	p -value, H_0 : c_2 subsample =				
	c_2 full sample - subsample			0.04	0.67

T-statistics are in parentheses. ** Significant at 1 percent level. * Significant at 5 percent level. Variance-covariance matrix estimated allowing for arbitrary correlations within state (clustered by state).

Table II: OLS Estimates, Dependent Variable: Log FERA Spending/Capita. This is the baseline spending table. The key rows are log share of households with radios and log voter turnout. Columns test the full specification, reduced missing-variable specification, competitive-state subsample, and rural-county subsample.

TABLE III
IV ESTIMATES, DEPENDENT VARIABLE: LOG FERA SPENDING/CAPITA

	I	II	III	IV
sample	full sample	competitive states	rural counties	full sample
instrumented variable	radio	radio	radio	turnout
c_1 : log share hhlds with radios	0.238 (1.0)	0.617 (1.9)	0.717* (2.3)	0.143* (2.7)
c_2 : log voter turnout	0.162** (2.9)	0.331** (2.8)	0.060 (0.8)	0.120 (1.9)
.	.	.	.	.
state effects	yes	yes	yes	yes
R^2	0.63	0.58	0.66	0.63
Number of observations	2490	1748	981	2470
F -stat, instruments, 1st stage	36	22	22	148
Overid restrictions, χ^2_{df} (p -value)	2.42 ₁ (0.12)	1.4 ₁ (.23)	0.43 ₁ (.51)	1.71 ₃ (0.64)
Hausman test for endogeneity, p -value	0.67	0.26	0.07	0.35

Voter turnout is instrumented by turnout and log turnout in the last gubernatorial election in each state, prior to 1933, and turnout in the 1928 presidential election. Radio is instrumented by ground conductivity in the county seat, and the land area share that is woodland. All independent variables in Table II, column I, are included but not displayed in the above regressions. *T*-statistics are in parentheses. ** Significant at 1 percent level. * Significant at 5 percent level.

Table III: IV Estimates, Dependent Variable: Log FERA Spending/Capita. Radio penetration is instrumented with ground conductivity and woodland share; voter turnout is separately instrumented using prior turnout. The table reports first-stage strength, overidentification tests, and Hausman tests.

7.4 Table IV: Estimated Effects on Turnout, Dependent Variable: Votes per Capita and Figure II: Total Estimated Effect of Radio on FERA Spending

TABLE IV
ESTIMATED EFFECTS ON TURNOUT, DEPENDENT VARIABLE: VOTES PER CAPITA

	I OLS 1920–1930	II IV 1920–1930	III OLS 1920–1940
share hhlds with radios (b_1)	0.117** (3.0)	0.200 (2.0)	0.073* (2.1)
radio*vote margin at state level	−0.932** (−3.5)	−1.329** (−4.0)	
vote margin at state level	−0.043 (−0.7)	0.027 (0.4)	−0.150* (2.3)
controls	yes	yes	yes
county fixed effects	yes	yes	yes
election year effects	yes	yes	yes
R^2	0.48	0.45	0.54
Number of observations	5740	5726	9007
Overid restrictions, χ^2_{df} (p-value)		1.14, (0.29)	
Hausman test for endogeneity, p-value		0.22	

In column II, radio is instrumented by ground conductivity in the county seat, and the land area share that is woodland. Controls in columns I and II include share illiterate, share aged 7–18 attending school, share 21+, share female, share urban, share black, share immigrants, log population. The regression in column III includes the same controls, excluding the share illiterate due to lack of data for 1940. T -statistics are in parentheses. ** Denotes significance at 1 percent level. * Denotes significance at 5 percent level. Variance-covariance matrix estimated allowing for arbitrary correlations within state (clustered by state).

(a) **Table IV: Estimated Effects on Turnout, Dependent Variable: Votes per Capita.** This table estimates whether radio increased voter turnout using county panel variation from 1920 to 1930 and, in one specification, 1920 to 1940.

Read:

- Table IV shows that radio penetration increased votes per capita. The OLS estimate for 1920-1930 is 0.117 and significant at the 1 percent level.
- The IV estimate is larger at 0.200, though less precisely estimated.
- The interaction between radio and the vote margin is negative, meaning radio matters less for turnout when elections are less competitive.
- Figure II reports a total effect of 0.61 percent higher FERA spending from a one percentage point increase in radio penetration.
- Most of the total effect, 0.54 percent, is direct; the indirect turnout channel is about $0.12 \times 0.57 \approx 0.07$ percent.

Detailed interpretation: The turnout table verifies one mechanism in the model: radio increases political participation. But the decomposition shows that turnout is not the whole story. The larger component is direct political accountability: radio users know who delivered or withheld relief and can reward or punish governors accordingly.

How it fits the paper: The table and figure together complete the mechanism. Radio changes voter behavior, but more importantly it changes the informational value of voters to politicians.

8 Short summary

- The paper studies whether media access affects government spending.
- The setting is the New Deal FERA program during the rapid diffusion of radio.

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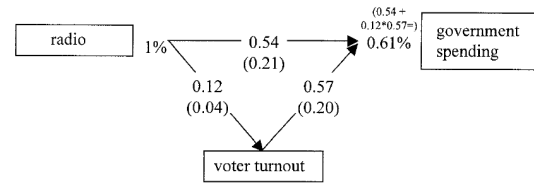


FIGURE II
Total Estimated Effect of Radio on FERA Spending

(b) **Figure II: Total Estimated Effect of Radio on FERA Spending.** The figure decomposes the effect of a one percentage point increase in radio penetration into a direct spending effect and an indirect turnout effect.

- The model predicts that governors allocate more relief to counties where voters are informed, turn out, and are electorally marginal.
- Empirically, counties with higher radio penetration receive more FERA relief per capita.
- The baseline OLS coefficient on radio is 0.138, and the effect is larger in competitive states and rural counties.
- IV estimates use ground conductivity and woodland share as instruments for radio penetration.
- Radio also increases voter turnout, but the larger effect on spending comes from direct information/accountability rather than turnout alone.
- Illiteracy is negatively related to spending, reinforcing the broader information mechanism.
- The results suggest that the distribution of political information affects the distribution of public resources.

9 Conclusion

9.1 Core conclusion

Strömberg shows that mass media can alter public spending by changing which voters are politically informed. In the 1930s, U.S. counties with more radio listeners received more New Deal relief funds. The result is statistically significant, economically meaningful, and consistent with a model in which politicians target voters who can observe and reward policy actions.

9.2 Economic intuition

A voter can only punish or reward a politician for a transfer if the voter knows who was responsible for the transfer. Radio made political information cheaper and more widely available. Counties with more radios therefore became more electorally valuable. Governors responded by allocating more FERA funds to those counties. Radio also raised turnout, but the paper's decomposition shows that the information/accountability channel is larger than the turnout channel.

9.3 Incremental contribution revisited

The paper's lasting contribution is to add information access to the political economy of redistribution. Classical models emphasize income, voting rights, or turnout. Strömberg shows that the political power of a group also depends on whether the group is informed. This distinction matters because the poor may have the right to vote but still be politically weak if they lack access to information about government responsibility.

9.4 Policy and research implications

The findings imply that new communication technologies can create winners and losers in the distribution of public goods. Radio improved the political position of some rural voters; television and the internet may have similar reallocative effects. The paper also suggests that media access is a component of democratic accountability. Policies that expand access to credible information can change not only beliefs and turnout but also the allocation of public resources.

9.5 Limitations

The analysis is historical and cross-sectional, and radio ownership is not randomly assigned. The IV strategy is persuasive but relies on exclusion restrictions: ground conductivity and woodland share must affect relief only through radio reception and ownership. The model also abstracts from endogenous media content,

newspaper competition, federal political incentives, and local lobbying. These limitations do not overturn the central evidence, but they guide how the estimates should be interpreted.

9.6 Takeaway

The key takeaway is that democratic influence is shaped by information, not just by formal voting rights. A mass medium that changes who knows what politicians do can change where government money goes.